

Coding & Solution

Date	2 JULY 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/anuradhadevara2-afk/OptiCrop
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn

Key Features Implemented	• Crop Recommendation using Machine Learning • Soil Nutrient Analysis (N, P, K) • Weather-Based Crop Prediction • Responsive Web Interface • Model Training using Random Forest Classifier • Input Validation • Prediction API using Flask
Pending / Incomplete Features	• Cloud Deployment • User Login System • Database Integration • Mobile Application
Setup / Run Instructions	1. Install Python 3.9+ 2. Install required libraries using pip install -r requirements.txt 3. Run python train.py to train the model. 4. Run python app.py to start the Flask server. 5. Open http://127.0.0.1:5000 in a web browser.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	yes
2	Meaningful variable and function names are used	yes
3	Code includes comments / documentation where necessary	yes
4	Error handling is implemented for critical operations	yes
5	The application runs without critical errors	yes
6	Code is committed to a version control repository	yes

Additional Notes / Comments:

The project successfully predicts the most suitable crop based on soil nutrients and climatic conditions using Machine Learning. The application follows a modular architecture with separate frontend, backend, machine learning model, and dataset components. It is easy to maintain, scalable, and can be extended with cloud deployment, real-time weather APIs, and mobile application support in future versions.